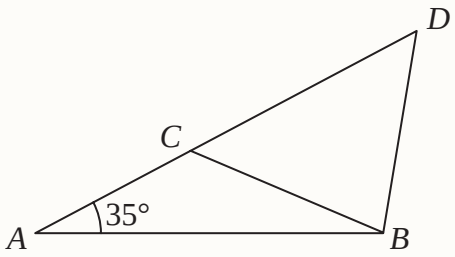
7.



Not to scale

Figure 3

Figure 3 shows the design for a structure used to support a roof.

The structure consists of four wooden beams, *AB*, *BD*, *BC* and *AD*.

Given $AB = 6.5\,\text{m}$, $BC = BD = 4.7\,\text{m}$ and angle $BAC = 35°$

(a) find, to one decimal place, the size of angle *ACB*,

(3)

(b) find, to the nearest metre, the total length of wood required to make this structure.

(3)